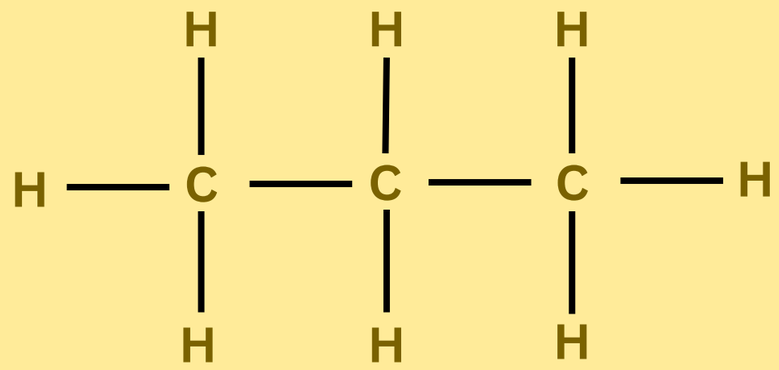
Propane

1

MOLECULAR FORMULA

C_3H_8

LEWIS STRUCTURE

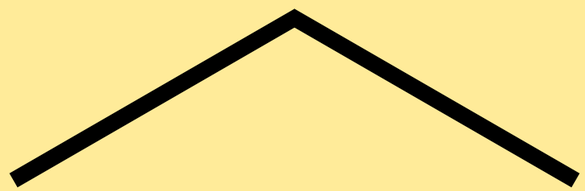


No lone pairs required

CONDENSED STRUCTURAL FORMULA

$CH_3CH_2CH_3$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

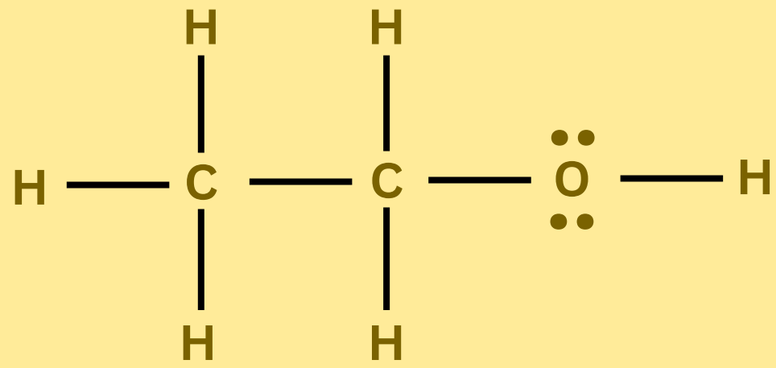
Ethanol

2

MOLECULAR FORMULA

C_2H_6O

LEWIS STRUCTURE



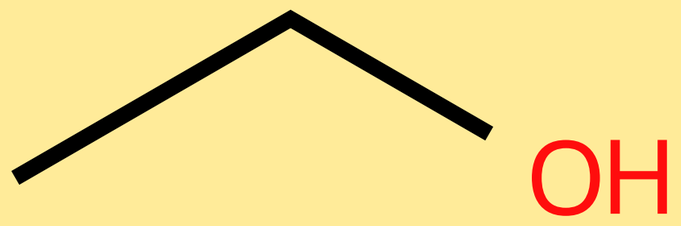
2 lone pairs on O; no lone pairs on C or H

CONDENSED STRUCTURAL FORMULA

CH_3CH_2OH

is also written as CH_3CH_2-OH

BOND-LINE FORMULA



VERIFICATION STRIP

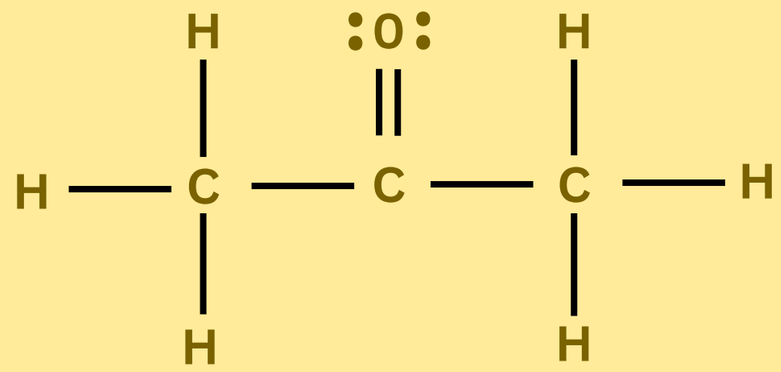
- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

Acetone [propan-2-one]

MOLECULAR FORMULA

C_3H_6O

LEWIS STRUCTURE

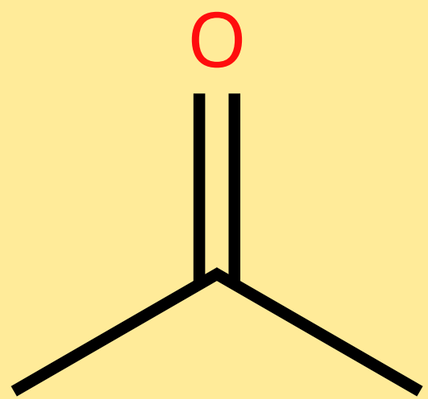


2 lone pairs on O; no lone pairs on C or H

CONDENSED STRUCTURAL FORMULA

CH_3COCH_3

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

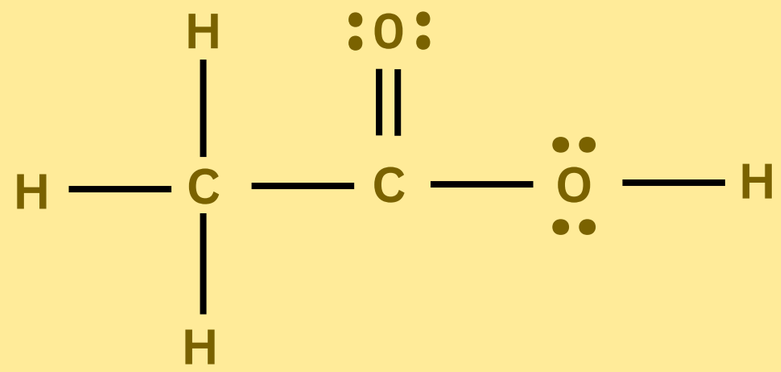
Acetic acid [ethanoic acid]

4

MOLECULAR FORMULA

$C_2H_4O_2$

LEWIS STRUCTURE



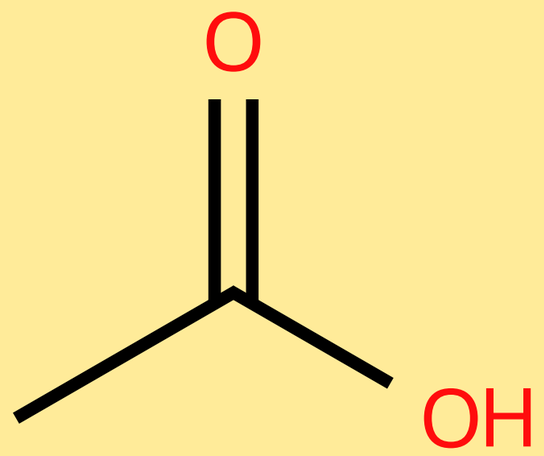
2 lone pairs on each O; none on C or H

CONDENSED STRUCTURAL FORMULA

CH_3COOH

is also written as $CH_3C(=O)OH$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

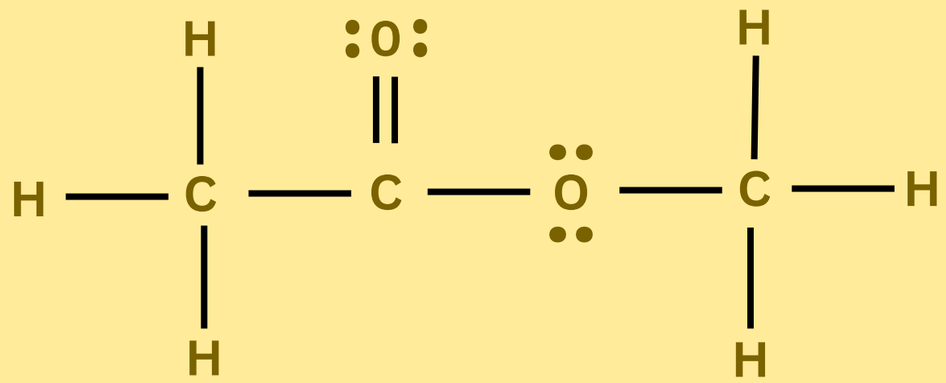
Ethyl acetate [ethyl ethanoate]

5

MOLECULAR FORMULA

$C_4H_8O_2$

LEWIS STRUCTURE



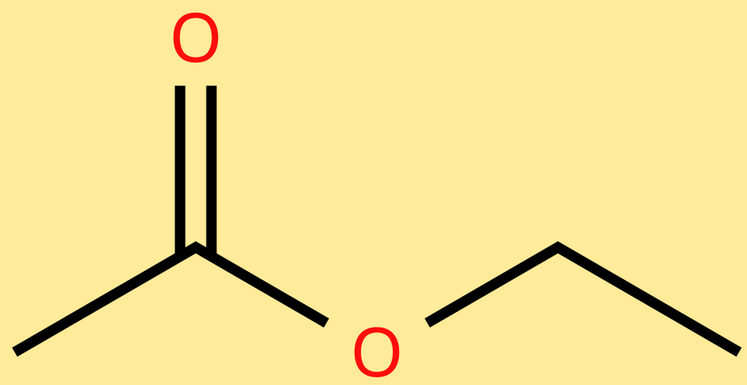
2 lone pairs on each O; none on C or H

CONDENSED STRUCTURAL FORMULA

$CH_3COOCH_2CH_3$

is also written as $CH_3C(=O)OCH_2CH_3$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

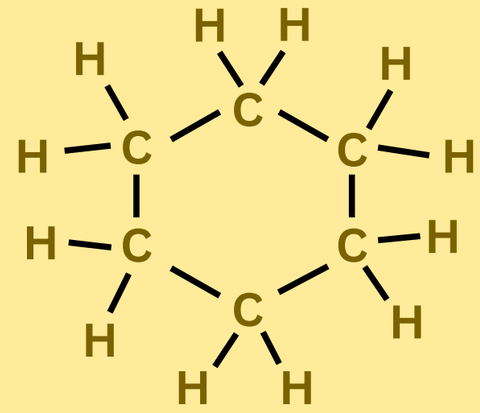
Cyclohexane

6

MOLECULAR FORMULA

C_6H_{12}

LEWIS STRUCTURE



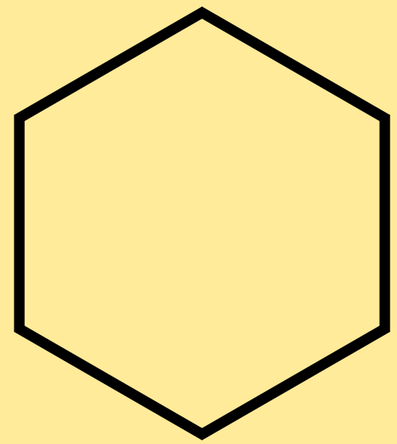
No lone pairs; every C has 2 ring bonds + 2 C-H bonds

CONDENSED STRUCTURAL FORMULA

$(CH_2)_6$

six CH_2 joined in a ring (cyclic)

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

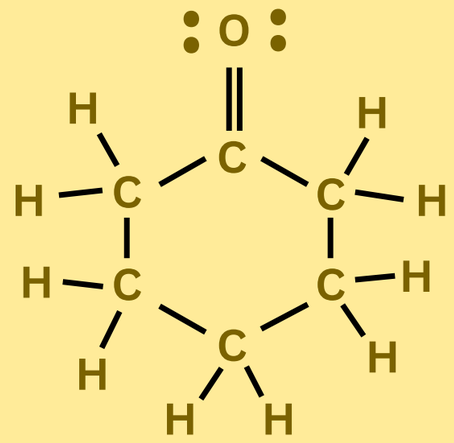
Cyclohexanone

7

MOLECULAR FORMULA

$C_6H_{10}O$

LEWIS STRUCTURE



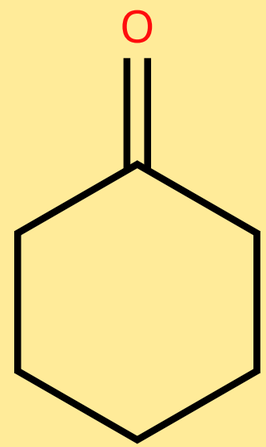
2 lone pairs on O;
none on C or H

CONDENSED STRUCTURAL FORMULA

$(CH_2)_5C=O$

five CH_2 joined + one C=O joined in a ring

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

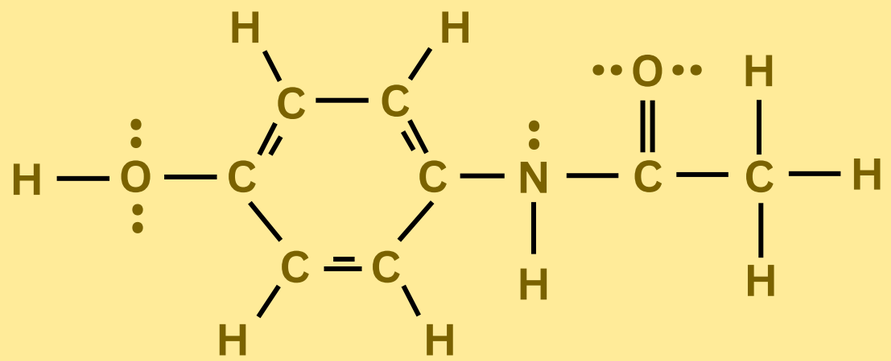
Acetaminophen [paracetamol]

8

MOLECULAR FORMULA

$C_8H_9NO_2$

LEWIS STRUCTURE



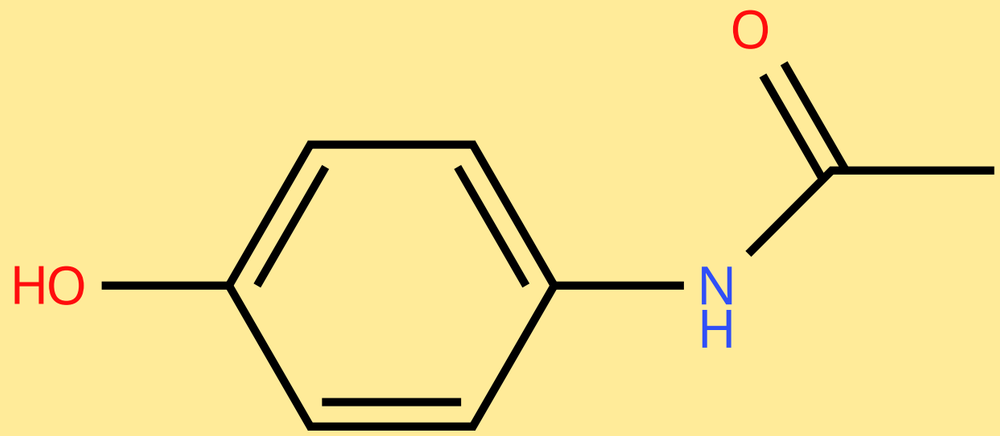
Lone pairs: 2 on each O, 1 on N; none on C or H

CONDENSED STRUCTURAL FORMULA

$CH_3CONHC_6H_4OH$

is also written as $CH_3C(=O)NH\text{-}C_6H_4\text{-}OH$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

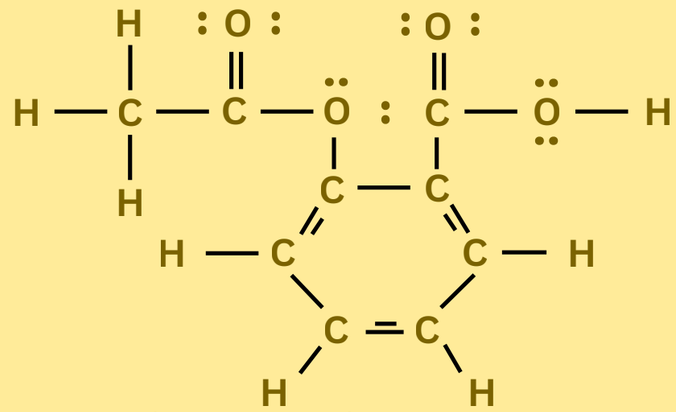
Aspirin [acetylsalicylic acid]

9

MOLECULAR FORMULA

$C_9H_8O_4$

LEWIS STRUCTURE



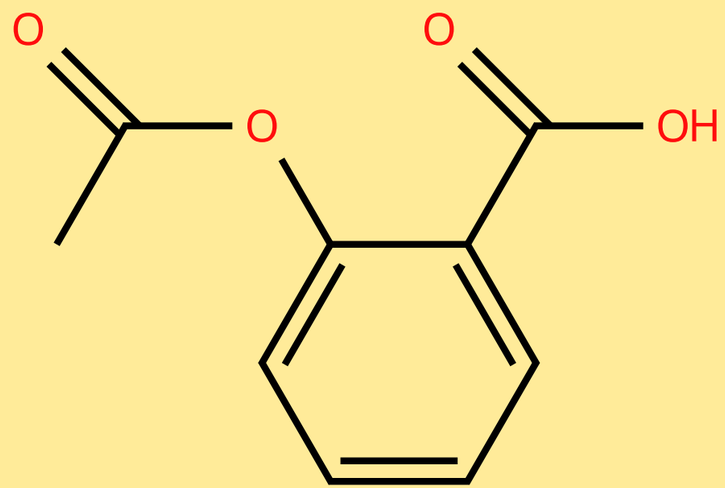
Lone pairs; 2 on each O; none on C or H

CONDENSED STRUCTURAL FORMULA

$CH_3CONHC_6H_4OH$

is also written as $CH_3C(=O)NH\text{-}C_6H_4\text{-}OH$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed

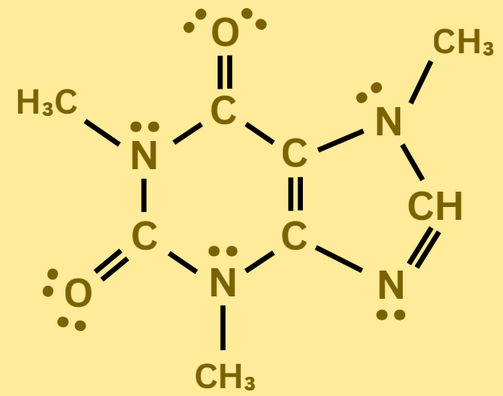
Caffeine

10

MOLECULAR FORMULA

$C_8H_{10}N_4O_2$

LEWIS STRUCTURE



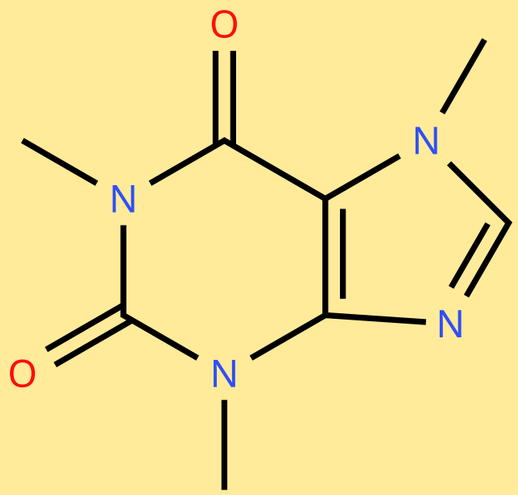
6 Lone pairs: 2 on each O; 1 on each N; none on C and H

CONDENSED STRUCTURAL FORMULA

$C_5HN_4O_2(CH_3)_3$

is also written as $CH_3C(=O)NH\text{-}C_6H_4\text{-}OH$

BOND-LINE FORMULA



VERIFICATION STRIP

- ☑ Atom count verified
- ☑ Valence checked
- ☑ Bond order checked
- ☑ Same connectivity confirmed